

Ans. to Q. No. 1(a)

$$G(s) = \frac{30}{s^2 + 4s + 20}$$

$$G(s) = \frac{K \omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

$$\omega_n^2 = 20$$

$$\omega_n = \sqrt{20} \text{ rad/s} = 4.47 \text{ rad/s}$$

↓
natural frequency

$$2\zeta \omega_n = 4$$

$$\zeta = \frac{4}{2\omega_n} = \frac{4}{2 \times \sqrt{20}} = 0.447$$

↓
damping ratio

Ans. to the Q. No. 1(b)

$$\zeta = 0.447 \text{ from 1(a).}$$

Since $0 < \zeta < 1$, the system is underdamped.
↓
damping ratio

Ans. to the Q. No. 1(c)

$$\% OS = e^{\frac{-\zeta \pi}{\sqrt{1-\zeta^2}}} \times 100\%$$

$$\zeta = 0.447 \text{ from 1(a)}$$

$$\therefore \% OS = e^{\frac{-0.447 \times 3.1416}{\sqrt{1-0.447^2}}} \times 100\%$$

$$= 0.2081 \times 100\%$$

$$= 20.81\%$$

$$\text{Settling time} = \frac{4}{\zeta \omega_n}$$

$$= \frac{4}{0.447 \times 4.47}$$

$$= 2.0019 \text{ s}$$

$\therefore \% OS = 20.81\%$ and settling time 2.0019 s .